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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING – (24CI3201)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	ModelGuard AI: Real Time Machine Learning Guardrail Gateway with Automated Input Validation	
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Abstract

ModelGuard AI is a real-time machine learning-based guardrail system designed to make AI applications safer, more reliable, and easier to monitor. It acts as an intelligent security layer between users and AI applications by checking incoming inputs before they are processed. The system identifies invalid, unusual, or potentially harmful requests and prevents inappropriate inputs from reaching the application, thereby improving the overall reliability of AI-based software systems.

The system uses FastAPI and Pydantic for API development and structured input validation, while Scikit-learn is used to detect unusual input patterns through machine learning techniques. MLflow is used to track and manage machine learning models, experiments, and performance results. The application is deployed using Docker and Kubernetes to support consistent and scalable execution. Prometheus and Grafana are used to monitor system performance and provide real-time visualization of important metrics. SonarQube and Trivy are incorporated to identify code quality issues and security vulnerabilities during the development and deployment process.

The project follows an adaptive software engineering approach by combining Agile, DevOps, SecOps, and MLOps practices. Agile practices can be used to manage requirements, user stories, backlogs, and iterative development. The system also supports continuous improvement by analyzing monitoring results and model performance to enhance its validation and detection capabilities over time. ModelGuard AI ultimately aims to provide a secure, adaptive, and well-monitored foundation for developing and deploying modern AI-powered applications.